```
        context.beginPath();
        geoPath(sampleData[x]);
        context.stroke();
        context.fill();
      }
    };
};
```

Draws each triangle to canvas

You can see some key differences between listings 11.6 and 11.5. In contrast with SVG, where you can move elements around as well as redraw them, you always have to clear and redraw the canvas to update it. Although it seems this would be slower, performance increases on all browsers, especially those that don't have the best SVG performance, because you don't need to manage hundreds or thousands of DOM elements. The graphical results, as seen in figure 11.4, demonstrate that it's hard to see the difference between SVG and canvas rendering.

11.1.3 Mixed-mode rendering techniques

The drawback with using canvas is that you can't easily provide the level of interactivity you may want for your data visualization. Typically, you draw your interactive elements with SVG and your large datasets with canvas. If we assume that the countries we're drawing aren't going to provide any interactivity, but the triangles will, then we can render the triangles as SVG and render the countries as canvas using the code in the following listing. This requires that we initialize two versions of `d3.geo.path`—one for drawing SVG and one for drawing canvas—and then we use both in our zoomed function.

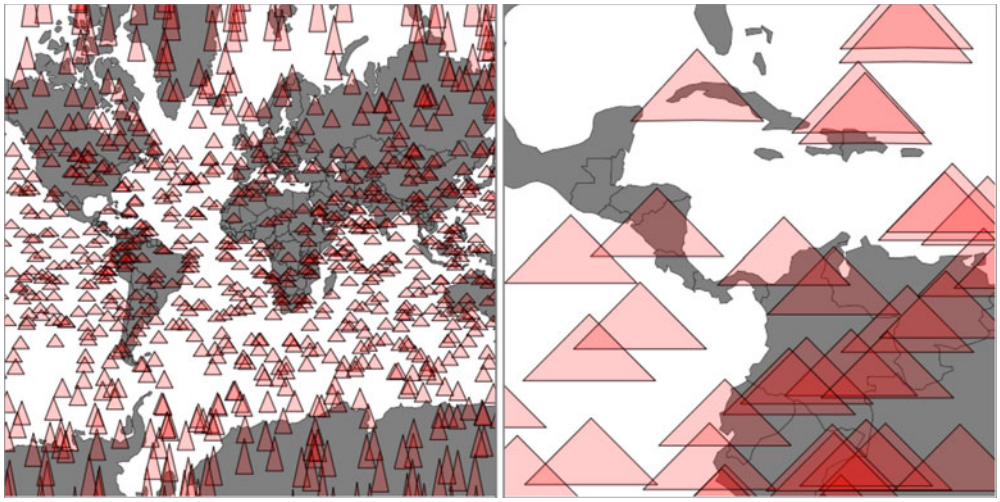


Figure 11.4 Drawing our map with canvas produces higher performance, but slightly less crisp graphics. On the left, it may seem like the triangles are as smoothly rendered as the earlier SVG triangles, but if you zoom in as we've done on the right, you can start to see clearly the slightly pixelated canvas rendering.